When Simulation-Based Sample Supplementation Fails in Cross-Domain Bearing Fault Diagnosis: The Recording-Fingerprint Effect and an Evaluation Protocol That Reveals It

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Abstract

Physics-based simulation, notably "digital twin" generators, is a popular source of synthetic fault samples for cross-domain bearing fault diagnosis when labeled target data are scarce. On the standard CWRU 12 kHz drive-end benchmark it is *structurally* unable to supply class information. Every "fault type × load" cell is a *single recording*, and the class-conditional magnitude spectrum is a stable property of it: a class's spectral envelope recurs across the four loads at cosine 0.944, yet is not predictable from fault physics, since a source-domain envelope transfers to the target at 0.924 while one estimated from five labeled *target* samples reaches 0.995. Class identity is thus a per-bearing *recording fingerprint*. A calibrated McFadden–Smith twin produces nine classes with near-identical spectra and, trained on twin data alone, reaches macro-F1 0.123 (chance 0.10). Its few-shot protocol compounds this: support and test windows share the fingerprint, so cross-recording evaluation lowers reported macro-F1 by 0.15–0.19 (up to 0.31) and equalizes augmentation-based methods with the trivial baseline. Coloring white noise with the target envelope is the most robust augmentation when the target recording is atypical; adding a physical impulse train degrades fi-

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delity and accuracy. On Paderborn, where several bearings share each class, class-mean spectra remain indistinguishable (cosine 1.000) although classes are defined by damage location; there same-recording optimism of +0.386 splits into session (+0.161) and specimen (+0.225) terms. The advantage of fingerprint calibration over raw physics inverts between the benchmarks.

Keywords: bearing fault diagnosis, benchmark evaluation, cross-domain transfer, digital twin, few-shot learning, simulation-based data augmentation, spectral fingerprint

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1. Introduction

Deep models for bearing fault diagnosis are notoriously fragile under domain shift, and the scarcity of labeled data on a newly commissioned machine makes transfer learning unavoidable. Two families of remedies dominate the literature: (i) unsupervised domain adaptation, typically adversarial [1] or moment-matching [2], and (ii) the generation of additional labeled samples for the target operating condition, either by physics-based simulation [3, 4, 5] or by generative models [6].

The simulation route is attractive because a physical model is, in principle, transferable [7, 8]: bearing characteristic frequencies scale with shaft speed, so a model of a spalled outer race at 1797 rpm can be re-instantiated at 1730 rpm without any new data. Many recent papers report that augmenting training with simulated faults improves cross-domain accuracy [5, 9, 10, 11, 3, 4].

This paper asks a prior question that the field has largely skipped: *what class-related information can a physical simulator actually supply on the benchmarks we use to test it?* We answer it for the most widely used benchmark, the CWRU 12 kHz drive-end dataset, and the answer is uncomfortable: on this benchmark the simulator supplies essentially nothing, for a structural reason, and the experimental protocol used to evaluate such methods is itself inflating results.

1.1. Contributions

- **C1 (diagnosis).** A physically calibrated twin remains 99.99% separable from real data under a domain discriminator, and a classifier trained only on twin data transfers to the real target domain at macro-F1 = 0.123, i.e. chance for ten classes (Section 4).

- **C2 (root cause).** Class identity in CWRU is dominated by a *recording fingerprint*. Per-class spectral envelopes are stable across loads (cosine 0.944 averaged over all load pairs), but the source-domain envelope predicts the target-domain envelope only to 0.924. The twin reproduces neither (Section 5).

- **C3 (protocol).** The few-shot evaluation protocol draws support and test windows from the same recording, so both share the fingerprint. Under a cross-recording protocol the reported macro-F1 drops by 0.15–0.19 (average over four support domains; up to 0.31), and augmentation-based methods become indistinguishable from training on the labeled support set alone (Section 6). We recommend this control for future work.

- **C4 (constructive).** Estimating the per-class spectral envelope from k labeled target samples is accurate (0.995 cosine at $k = 5$), and coloring white noise with it is the most robust augmentation when the target recording is atypical. Adding a physical impulse train on top of it *degrades* both domain fidelity and accuracy, consistently across settings (Sections 8 and 9).

- **C5 (side finding).** Load 3 of CWRU is an outlier operating condition: label-free transfer among loads 0–2 exceeds 0.98, while any transfer involving load 3 falls to 0.64–0.76. Conclusions drawn from the popular S0→S3 direction alone do not generalize (Section 6).

- **C6 (cross-benchmark).** On Paderborn, where several independent bearings share each damage class, class-mean magnitude spectra of healthy, inner-ring and outer-ring bearings are identical (cosine 1.000; CWRU: 0.626), and remain so after the dominant rig tone is filtered out (margin +0.003). The CWRU class separation is therefore a property of the individual specimens, not of the classes. Same-recording evaluation is optimistic by +0.386 macro-F1 there; because each Paderborn bearing was measured 20 times, that optimism can be split by re-evaluating on a different *recording of the same bearings*, which gives +0.161 for the session and a further +0.225 for the change of specimen, and it survives removal of the rig tone that holds 93% of the spectral energy (+0.321 remains). A twin whose physics exactly matches the class definition is

still the worst augmentation tried. The advantage of fingerprint calibration over raw physics inverts between CWRU and Paderborn in the direction the mechanism predicts, which serves as a positive control (Section 7).

2. Related Work

Cross-domain and few-shot bearing diagnosis. The problem setting we study is standard: a model trained under one operating condition must be reused under another, and labels are scarce on the target [12, 13]. The dominant response is unsupervised domain adaptation, usually adversarial, which aligns source and target feature distributions without target labels [1, 14, 15]. A complementary line attacks label scarcity directly through meta-learning and metric-based few-shot schemes [16], while the backbone architectures themselves are now well surveyed [17]. Both are typically evaluated on CWRU or a handful of sibling benchmarks, and both depend on the support set being representative of the target test distribution. We show that under the usual window-level split this assumption is violated in a specific and quantifiable way.

Simulation and digital-twin sample supplementation. Because a physical model transfers by construction, generating labeled fault samples by simulation is attractive precisely when target labels are missing. Recent work builds bearing dynamics models and uses the resulting *twin data* as a labeled source: partial domain adaptation driven by a digital twin [10], discriminative graph learning that transfers knowledge from simulated to real signals [9], multimode collaborative transfer learning on twin samples [5], and digital-twin-assisted domain adversarial frameworks for machinery such as centrifugal pumps [11]. A recurring theme in this literature is the simulation-to-reality gap, and much of the machinery — physics-enhanced translation between simulated and measured signals [3], generative alignment of the two distributions [4] — exists to close it. The same gap motivates domain randomization in robotics [18], where the simulator is deliberately not made faithful and the burden shifts entirely onto the augmentation distribution.

Our contribution is orthogonal to closing that gap. We ask what class-discriminative information the simulated data can carry in the first place, given the structure of the benchmark, and we answer it with a negative result

that no amount of gap-closing can repair: on CWRU the class-conditional spectrum is a property of the individual recording rather than of the fault, so a simulator with perfect fidelity to the physics still cannot supply the contrast that separates the classes.

Surrogate data. Our phase-randomization control follows the standard surrogate-data procedure for testing whether a statistic depends on structure beyond the power spectrum [19, 20]. We use it in the reverse direction from the usual application: rather than asking whether a signal is nonlinear, we use phase-randomized surrogates as a deliberately *structure-free* augmentation baseline, to separate the contribution of the spectral envelope from that of any temporal structure.

3. Data, Taxonomy and Protocol

3.1. CWRU and the ten-class taxonomy

We use the CWRU 12 kHz drive-end recordings [21]. The conventional four-class taxonomy (normal, ball, inner race, outer race) is saturated on cross-load transfer — a source-only model reaches 100% accuracy in all twelve transfer directions — so we adopt a ten-class taxonomy: $\{\textsc{Normal}\} \cup \{B, IR, OR\} \times \{0.007'', 0.014'', 0.021''\}$, using the official files for the four loads $\{0, 1, 2, 3\}$ (1730–1797 rpm). Windows of 1024 samples with stride 512 are z-scored individually.[1]

3.2. Few-shot split, and the defect we exploit later

For a target load, each class is split in time order: the last 40% of windows form the held-out test set; from the first 60%, k windows per class are drawn (with the seed) as the labeled *support* set, and the rest form an unlabeled *pool*. Crucially, the support and test windows come from the *same recording*. Section 6 shows this is not a detail.

[1] Every spectral scalar reported in this paper is reproduced, from the raw data, by one script released with the code (`tools/probe_spectral_numbers.py`). We make the point explicitly because a claim of this paper is that numbers reported in this area do not always survive re-derivation, and that claim would be worthless if our own did not.

3.3. Classifier and training

Every configuration is trained with the same model, so that each comparison isolates the training set rather than the architecture. The input is the raw time-domain window, z-scored sample-wise, and no spectral transform is applied before the classifier: the spectral statistics of Sections 4 and 5 are diagnostics on the data, not features of the model. The backbone is a one-dimensional residual network with a 7-tap convolution (stride 2, 32 channels), batch normalisation and ReLU, three residual blocks of two 3-tap convolutions each (32, 64 and 128 channels, stride 2 from the second block), global average pooling, and a linear projection to 64 features. One linear layer maps those features to class scores, and the domain discriminator used by the adversarial methods is a two-hidden-layer MLP of width 64 on the same features. The backbone has 118 368 parameters and the model 127 403 including the discriminator. Training minimises cross-entropy, plus a binary domain loss where a discriminator is used, with Adam at a learning rate of 10^{-3}, batch size 64 and 60 epochs, 200 for the Paderborn comparisons of Section 7; synthetic sets are generated with 400 windows per class. One seed controls the few-shot draw, the synthetic sample generation and the weight initialisation jointly.

4. Diagnosis I: What the Twin Supplies

4.1. The generator

We implement a McFadden–Smith style impulse-response twin [22] with SKF 6205-2RS geometry, whose ball-pass frequencies are computed from the shaft speed. The generator is calibrated against real CWRU spectra: the resonance range, the resonance bandwidth per defect size, the background-noise colouring, and the impulse-to-harmonic power ratio were all fitted so that synthetic band-energy shares match measured ones. In particular the synthetic peak envelope spectrum lands on the correct BPFO, so the twin is *not* an uncalibrated straw man.

4.2. Twin data are a different domain, and carry no class information

Training only on twin data and testing on the real target load (Table 1) gives macro-F1 $= 0.123 \pm 0.030$ against a chance level of 0.10. A linear domain discriminator on backbone features separates twin from real with accuracy 0.9999. The same table shows the ordering that the rest of the paper is

Table 1: Training only on a synthetic source, testing on the real target load (load 3, ten classes). Each entry is 20 runs — ten seeds, the whole configuration repeated twice — and the ± is the standard deviation over those 20 runs. "Disc." is a domain discriminator trained on backbone features to separate the synthetic source from real target data (0.5 = indistinguishable). The repeat is not padding: the two repetitions of the target-fingerprint row differ by 0.033 in the mean of ten seeds, and a single ten-seed draw would understate that row's spread by a factor of two.

Synthetic source	macro-F1	Disc.
Physics twin (source speed)	0.123 ± 0.030	0.9999
Source fingerprint + white noise	0.587 ± 0.045	0.9972
Source fingerprint + impulse train	0.491 ± 0.041	0.9994
Target fingerprint + impulse train	0.651 ± 0.056	0.9786
Target fingerprint + impulse (comb-free)	0.524 ± 0.061	0.9999
Target fingerprint + white noise	$\mathbf{0.910 \pm 0.077}$	**0.9337**

about: the target-calibrated envelope reaches 0.910, and every attempt to put physical structure back on top of it *lowers* the score.

5. Diagnosis II: the Recording Fingerprint

5.1. Class differences are stable across loads

Taking the smoothed magnitude envelope $A_c(f)$ as each class's spectral signature (Section 8), the signature of load 0 correlates with those of loads 1–3 at cosine 0.954, 0.939 and 0.924 respectively (mean 0.939); averaged over all six load pairs the value is 0.944. Class differences therefore survive a change of load and are a stable property of the *recording*, which is why real-data methods work well here. Table 4 reports a separate, unsmoothed statistic (Section 7) and its numbers are systematically lower for that reason alone.

5.2. But they are not predictable from fault physics

The nine synthetic fault classes have near-identical mean spectra (Fig. 2(b)): all occupy 48–64% (mean 55%) of their energy in the 2800–3300 Hz band, whereas the nine real classes at load 0 spread from 9% (OR014) to 67% (IR021), mean 32%. Table 2 quantifies this with the mean pairwise cosine distance between class-mean spectra: real data 0.374, twin 0.167.

Table 2: Mean pairwise cosine distance between class-mean magnitude spectra (higher = classes better separated). Target load 3.

Source	Mean inter-class distance
Real target (load 3)	0.374
Real source (load 0)	0.374
Physics twin	0.167
Target fingerprint + white noise	0.315
Target fingerprint + impulse train	0.346
Target fingerprint (comb-free) + imp.	0.319

Table 2 also exposes a subtlety that we return to in Section 9: the physics impulse train *increases* inter-class distance relative to the fingerprint-only condition, yet it transfers worse. The physical structure does not destroy class information; it pushes the sample distribution away from the real one.

5.3. Why physics cannot close the gap

Each CWRU fault file was recorded with a *different* damaged bearing, and the same bearing was re-used across loads for a given file. "What this class looks like" is therefore a property of that particular bearing and recording session. A simulator cannot know which bearing a target recording used. The fingerprint is estimable from target labels (Section 8) but not derivable from fault mechanics. In the terminology of the shortcut-learning literature [23], the classifier is free to learn the recording as the shortcut to the label, and on this benchmark the shortcut is not merely easier than the intended solution — it is the only signal present at all.

6. Evaluation Protocol: Same-Recording Leakage

The support and test windows of the standard few-shot split come from one recording, so they share the fingerprint. This is a leakage of the kind that has been catalogued as a general threat to machine-learning-based science [24]: the split is drawn at the window level while the statistical unit that carries the label is the recording, so the test set is not independent of the training set in the sense that matters. To quantify the resulting optimism we evaluate every configuration twice: on the held-out windows of the *same*

Table 3: Same-recording versus cross-recording evaluation ($k = 5$, macro-F1). Each cell is 6 runs — three seeds, the whole configuration repeated twice. Each block trains on the labeled support of one load and evaluates on its own held-out windows (*same*) and on the other three loads (*cross*). "Surrogate" = Fourier phase randomization of each support sample; "true aug" = 400 real labeled samples per class drawn from the unlabeled pool.

Support load	Training configuration	Same	Cross mean	Cross (to load 3)	Optimism
load 0	support only	0.987	0.868	0.696	+0.119
	+ fingerprint aug.	0.985	0.837	0.709	+0.148
	+ phase surrogate	0.989	0.826	0.737	+0.163
	+ true samples	1.000	0.864	0.695	+0.135
load 1	support only	0.988	0.871	0.684	+0.117
	+ fingerprint aug.	0.988	0.837	0.691	+0.151
	+ phase surrogate	0.992	0.811	0.661	+0.181
	+ true samples	1.000	0.894	0.714	+0.106
load 2	support only	0.999	0.871	0.767	+0.127
	+ fingerprint aug.	0.946	0.847	0.733	+0.099
	+ phase surrogate	0.798	0.684	0.596	+0.114
	+ true samples	1.000	0.929	0.881	+0.071
load 3	support only	0.998	0.725	—	+0.273
	fingerprint aug.	0.995	**0.809**	—	+0.186
	+ phase surrogate	0.997	0.686	—	+0.311
	+ true samples	1.000	0.688	—	+0.312
Mean over the four support loads					
	support only	0.993	0.834		+0.159
	fingerprint aug.	0.979	0.832		+0.146
	phase surrogate	0.944	0.752		+0.192
	true samples	1.000	0.844		+0.156

load, and on the *other* three loads, which are different recordings. Table 3 reports the result for $k = 5$.

Four conclusions follow.

Same-recording evaluation is optimistic by 0.15–0.19. Support-only training scores 0.987–0.999 on its own held-out windows but only 0.725–0.871 on other recordings (Fig. 1). Any reported few-shot result on this benchmark that uses the standard split should be read as containing this optimism unless a cross-recording control is provided.

No augmentation reliably beats the labeled support set. Averaged over the four support loads, cross-recording macro-F1 is 0.834 for support only, 0.832

for fingerprint augmentation, 0.844 for adding 400 real labeled samples per class, and 0.752 for phase surrogates. The three former are within noise of each other. In particular, adding large amounts of *real* data from the same operating condition does not improve cross-recording generalization: the bottleneck is the operating condition, not the sample count.

Fingerprint augmentation is a regularizer. Its advantage appears only when the support recording is atypical. With load 3 as the support load it reaches 0.809 against 0.725 for support only; with loads 0–2 as the support load it is worse by 0.02–0.03 (0.837 against 0.868 on load 0). The mechanism is that averaging the k support spectra into a class-level model suppresses recording-specific idiosyncrasies.

Load 3 is an outlier condition. A purely source-trained model, using no target labels at all, transfers among loads 0–2 at 0.93–1.00 but reaches only 0.71 when load 3 is the test recording; conversely, labeled support from loads 0–2 transfers into load 3 at only 0.68–0.77. The widely used S0→S3 direction is hard mainly because load 3 is peculiar; results reported on that direction alone should not be generalized.

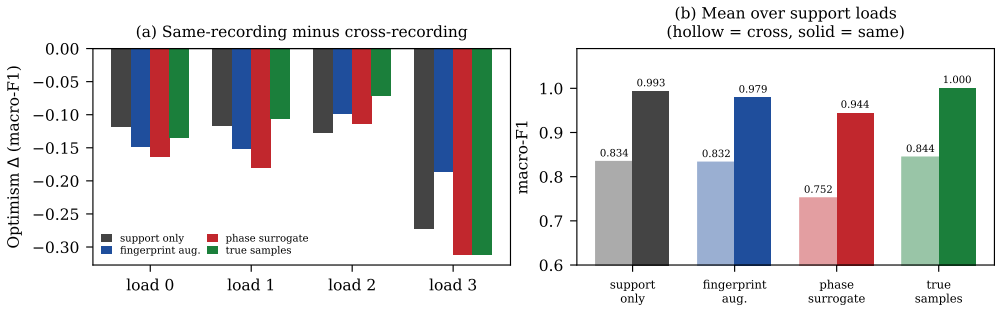


Figure 1: Same-recording versus cross-recording evaluation ($k = 5$, three seeds, two repetitions). (a) The optimism Δ of each training configuration, per support load: every bar is positive, i.e. every configuration looks better on its own held-out windows than on other recordings, and the effect is largest exactly where the literature's favourite direction (load 3) is involved. (b) Averaged over the four support loads. Hollow bars are cross-recording, solid bars same-recording. No augmentation reliably beats training on the labeled support set once the evaluation is cross-recording, and adding 400 *real* labeled samples per class (0.844) is within noise of the trivial baseline (0.834).

7. Cross-Benchmark Control: Paderborn

CWRU cannot falsify the fingerprint explanation, because it does not contain the control that would do so: every class has exactly *one* specimen. We therefore repeat the essential measurements on the Paderborn (KAt) dataset [25], in which several independent bearings share each damage class, measured under four operating conditions at 64 kHz.

We use 21 bearings: 6 healthy (K001–K006), 9 artificially damaged (KA series) and 6 damaged by accelerated lifetime tests (KI series). Classes are defined by *damage location*, read from the damage profile that ships with each bearing (outer ring OR, inner ring IR, healthy) — i.e. exactly the kinematic distinction that a fault model can express. Windows are 4096 samples (64 ms); at 64 kHz a 1024-sample window spans only 16 ms, which is shorter than one BPFO period at 1500 rpm, and we find that all bearings then produce indistinguishable spectra.

One caveat is recorded here rather than buried: in the subset we downloaded, every KA bearing is outer-ring and every KI bearing is inner-ring, so damage location and damage origin (artificial versus run-to-failure) are confounded. This does not affect the finding below, where the result is the *absence* of any class-conditional spectral structure, but it would matter for a claim about outer versus inner race specifically.

7.1. Class-conditional spectra are a rig property, not a class property

Table 4 compares the two benchmarks with one identical statistic: the cosine similarity between class-mean magnitude spectra, shown directly in Fig. 2.

On CWRU the ten classes are strongly separated in the magnitude spectrum (cosine 0.626). On Paderborn, where a class contains several independently manufactured bearings, averaging over specimens leaves *no* separation at all: within-class and between-class cosines are both 1.000 to three decimals. The spectra are dominated by a rig tone: at 1500 rpm the largest spectral line is at 100 Hz — exactly four times the shaft rate — in *every one* of the 21 bearings (measured on the full records it falls in 99.6–100.1 Hz for all of them), healthy, outer-ring and inner-ring alike. In a 4096-point window that line lands in the bin at 93.75 Hz and *that single bin* carries 54% of the total spectral energy (53.97–54.31% across the 21 bearings); the line together with two neighbouring bins on each side carries 99.5–99.8%.

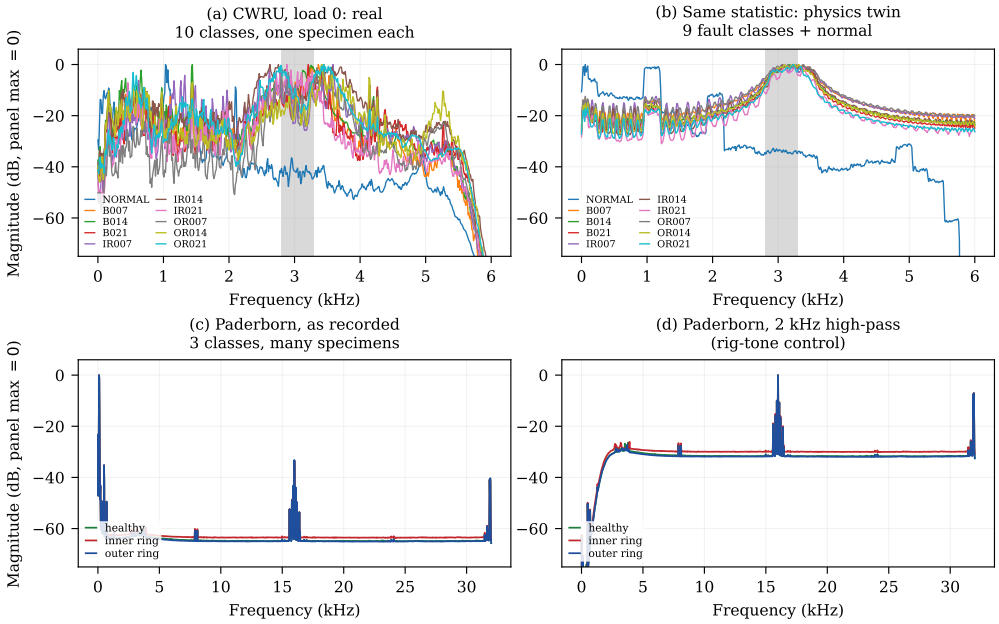


Figure 2: Class-mean magnitude spectra under the identical, unsmoothed statistic of Table 4. Each panel is normalized to its own maximum (0 dB); all four share the same −75 dB floor, the same linear frequency axis and the same per-class colours, and the shaded span marks the 2800–3300 Hz band whose energy fraction is quoted in Section 5. (a) CWRU, ten classes, one physical specimen each: visibly separated. (b) The physics twin, same statistic: the nine fault classes collapse into one band. Their median pairwise separation is 2.5 dB (minimum 0.9 dB), against 6.7 dB (minimum 3.7 dB) for the nine real classes, and across frequency the nine twin curves span a median of 6.5 dB against 18.8 dB. This is the whole negative result in one panel — the simulator produces nine faults whose mean spectra are interchangeable, so no classifier trained on them can carry the contrast that separates the real classes. (c) Paderborn as recorded: the 100 Hz rig line of Section 7 lands in the 93.75 Hz bin (with leakage into its neighbour at 109 Hz) and carries essentially all the energy, and the three damage classes are indistinguishable because everything else lies 60 dB below it. (d) The same recordings after a 2 kHz high-pass removes that line — the objection that one dominant tone is flattening the comparison. The 16 kHz rig resonance now dominates, yet the three class means still track each other to within a few dB across the whole band: the absence of class structure is not an artefact of the tone. Compare (d) with (a), where the same statistic separates the real classes by 10–30 dB.

Table 4: Class-mean magnitude spectra, identical statistic on both benchmarks: the *unsmoothed* mean magnitude spectrum of each class, so the two benchmarks are directly comparable. This is deliberately not the smoothed envelope $A_c(f)$ of Section 8 used elsewhere, which is smoother and yields higher cosines. "Within" = different specimens, same class; "between" = different specimens, different class. CWRU has one specimen per class, so only the between-class numbers exist there; its "same class across loads" row is the closest analogue of a within-class measurement. The last two rows remove the rig tone of Section 7 by high-pass filtering at 2 kHz, which is the control for the objection that one dominant line could flatten everything.

Benchmark	Quantity	Cosine	Margin
CWRU (one specimen per class)			
	between class, load 0	0.626	—
	between class, load 3	0.626	—
	same class, load 0 vs 1	0.870	—
Paderborn (21 specimens)			
	within class	1.000	
	between class	1.000	+0.000
Paderborn, high-pass 2 kHz			
	within class	0.992	
	between class	0.989	+0.003

That single tone invites an obvious objection to the claim above: of course the class means coincide, if one line holds almost all the energy. Filtering it away answers the objection. After high-pass filtering at 2 kHz the within-class cosine falls to 0.992 and the between-class cosine to 0.989; the margin moves from 0.000 to +0.003. Removing the rig tone does not reveal a hidden class signature — there is none to reveal in the magnitude spectrum — which is the opposite of what happens on CWRU, where the between-class cosine is 0.626 in every load. Filtering before *training* rather than before the statistic gives the same answer, and Section 7.4 reports that control.

The two facts together identify the mechanism. The class separation that makes CWRU look easy is not a property of the fault classes; it is a property of the individual specimens and recordings that happen to carry the class labels. Where several specimens share a label, the separation disappears.

Table 5: Same-recording versus bearing-disjoint evaluation on Paderborn (condition N15_M07_F10, three classes, 5 training bearings per class, 200 epochs, ten seeds, macro-F1). The training set is files 1–4 of the training bearings; the same-recording test set is its held-out 30% of windows. All ten runs are included, including the one that did not converge (see the text below).

	Same recording	Bearing-disjoint	Optimism
Mean over 10 seeds	0.891	0.505	+0.386
Std. deviation	0.120	0.161	
CWRU, for comparison (mean over four support loads)			
Support only	0.993	0.834	+0.159

7.2. Same-recording leakage reproduces on a second benchmark

If the optimism of Section 6 were an artifact of CWRU, it would not appear here. We train one model per seed on five bearings per class and evaluate it twice: on held-out windows drawn from the *same* bearings (30% of the training recordings), and on one bearing per class that the model has never seen. Both test sets are class-balanced and the metric is macro-F1. They are the two endpoints that Table 6 separates, from that same run.

The optimism is +0.386, well above the CWRU average and above CWRU's worst case. Same-recording evaluation is not a quirk of one benchmark; it is a consequence of drawing support and test windows from one recording, and it reproduces wherever a benchmark is organized as a small number of long recordings. We also note that under a bearing-disjoint protocol the features do retain *some* transferable signal (0.505 against 0.333 chance), although the optimism accounts for more than two thirds of the performance above chance (0.386 of 0.558).

The spread is larger here than on CWRU, and one of the ten runs deserves a comment. Seed 6 failed to converge: its macro-F1 on the *training* set is 0.60 where the other nine reach 0.98–1.00, and all three of its test scores are correspondingly low. Re-running the identical configuration converged (1.00 on the training set, $L1 = 0.927$), so this is a transient optimization failure rather than a property of the split. We report it rather than replace it: the contrast is measured *within* a run, and its optimism, +0.12, lies within the range of the other nine (+0.13 to +0.66). Discarding it would have raised the same-recording mean from 0.891 to 0.931.

Table 6: Decomposing the optimism, and the same decomposition with the rig tone removed. One training set (files 1–4 of five bearings per class, first 70% of windows) is evaluated on three test sets that differ by exactly one factor each: L2 moves only the recording session, L3 moves the specimen as well. Same condition, epochs and seeds as Table 5; the *no filter* column is that table's two endpoints. In the last two columns every window is masked in the frequency domain and renormalised before training, and the whole procedure is repeated from scratch: the notch removes 100 Hz and its first twenty harmonics with ±2 bins each, and the high-pass is the 2 kHz control of Table 4.

	No filter	Notch	High-pass
Spectral energy removed	—	93.3%	99.7%
L1 same file, held-out windows	0.891	0.899	0.908
L2 files 5–8, same bearings	0.730	0.772	0.776
L3 held-out bearings	0.505	0.577	0.550
Session term (L1−L2)	+0.161	+0.127	+0.132
Specimen term (L2−L3)	+0.225	+0.195	+0.226
Total optimism (L1−L3)	+0.386	+0.321	+0.358

7.3. The optimism decomposes into a session term and a specimen term

The two endpoints above leave the decisive question open: between them, *two* factors change at once. The bearing-disjoint test set is a different recording session *and* a different physical bearing. CWRU cannot separate them, because each class is a single specimen. Paderborn can, because each of the numbered bearings was measured 20 times under the same condition: a model can be trained on files 1–4 of a set of bearings and tested on files 5–8 of exactly those same bearings — a different recording session of the same specimen. The three test sets are drawn as a ladder in Fig. 3(a), where the drop between consecutive points is exactly one of the two terms computed below.

The result is unambiguous, and the specimen term is the larger of the two. Simply re-measuring the *same* bearings costs +0.161 macro-F1: that much of the optimism is pure session leakage, with the specimen held fixed. Changing the specimen on top of that costs a further +0.225. The two explanations that the CWRU protocol cannot distinguish — "the model keys on the recording" and "the model keys on the individual bearing" — are therefore both real, with the specimen carrying the larger share.

Two qualifications belong here. First, the specimen term is far more variable than the session term: dropping it has a standard deviation of 0.209 against 0.046 for the session drop, and one of ten seeds (seed 3, $L3 = 0.77$) even reverses the sign. Which bearing is held out matters more than which session it was measured in, as expected if the specimen term is carried by that bearing's individual spectral signature. The ten splits between them draw on 15 of the 21 bearings (five of the six healthy, four of the six inner-ring and six of the nine outer-ring), so what varies across seeds is a property of specimens in general rather than one unlucky draw.

Second, $L3 = 0.505$ is well above chance, so cross-specimen transfer is not zero: part of the fault physics does survive the change of specimen. Neither term rests on a handful of runs: over the ten seeds the session term is positive in all ten and the specimen term in nine, a two-sided sign test giving $p = 0.002$ and $p = 0.021$ without any normality assumption, and the 95% confidence intervals for the two means are +0.128 to +0.193 and +0.076 to +0.374. Nor is the decomposition a property of this one network: a ResNet with a quarter of the parameters (30 099 against 118 368) and the same recipe gives $L1 = 0.883$, $L2 = 0.727$, $L3 = 0.537$ — session +0.156, specimen +0.189, the same ordering and the same total to within 0.04 (its ten runs reach a training macro-F1 of 0.98 on average, two of them 0.88).

7.4. The decomposition is not carried by the rig tone

The 100 Hz line of Section 5 is a property of the test rig rather than of any bearing, so it cannot encode a specimen's identity. A reader may object that it is nearly all the classifier sees, so the decomposition measures little but the recovery of that one line. We test this by removing the line before training and repeating the whole decomposition with the same splits, seeds and schedule. Two filters are used: a notch at 100 Hz and its first twenty harmonics, ±2 bins each, which discards 93.3% of the total spectral energy, and the 2 kHz high-pass of Table 4, which discards 99.7%.

Both filters leave the conclusion of the section standing (Table 6). Same-recording evaluation moves little — L1 is 0.899 and 0.908 against 0.891 unfiltered — although more than nine tenths of the energy is gone. Total optimism is +0.321 and +0.358 against +0.386. The session term is the one that moves consistently: it falls to +0.127 and +0.132 from +0.161, a paired reduction of 0.034 ± 0.012 and 0.028 ± 0.011 standard error, and it is lower in nine of the ten seeds under each filter — the exception is the run of the previous subsection that did not converge. The specimen term moves by

-0.030 ± 0.018 under the notch and by $+0.001 \pm 0.023$ under the high-pass, i.e. in no direction that the two filters agree on, so we read it as unchanged. Roughly a fifth of the session leakage is therefore carried by the dominant line; the rest of it, and the whole of the specimen term, lie in the remainder of the spectrum — the bearing resonances and their sidebands, which is where the fault signature itself is.

That discarding more than 99% of the signal energy still leaves a same-recording optimism of 0.32–0.36 is the decisive form of this control: the effect is not an artefact of one conspicuous spectral line, and a protocol that guards against it cannot be defeated by notch-filtering the input. Fig. 3 draws the same numbers: in (a) the three ladders are nearly parallel, and in (b) the session bars fall under either filter while the specimen bars do not.

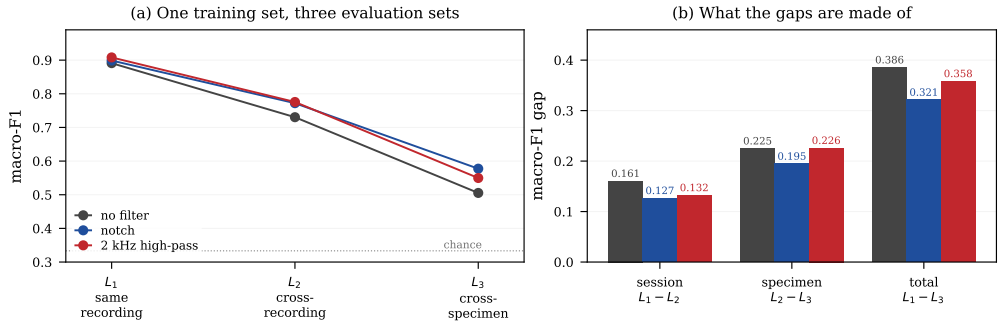


Figure 3: The decomposition of Table 6, and the same decomposition with the rig tone removed. (a) One training set, three evaluation sets that differ by one factor each; the drop between consecutive markers is exactly one of the terms plotted in (b). The three curves are the unfiltered data and the two filtered variants: removing 93.3% (notch) or 99.7% (high-pass) of the spectral energy moves any level by at most 0.07 and does not remove the decline. (b) The two terms and their sum, bars are means over ten seeds. Filtering reduces the session term from 0.161 to 0.127 and 0.132, roughly a fifth, and leaves the specimen term without a consistent direction (0.225 to 0.195 and 0.226), so the total stays in 0.32–0.36.

7.5. Where the kinematics match the labels

The Paderborn classes are defined by damage location, so here the impulse train at BPFO/BPFI is not merely plausible: it is the right physics for the label. We therefore repeat the no-source few-shot experiment of Section 9 with these classes, evaluating on held-out bearings.

Table 7: No-source few-shot augmentation on Paderborn, classes defined by damage location (healthy / inner ring / outer ring), bearing-disjoint test, macro-F1. Chance is 0.333. Five seeds at $k = 5$, three at $k = 20$ and $k = 50$. The last column is the same setting as Table 9 except that the class labels are, on this benchmark, consistent with the fault kinematics.

k (windows/class)	Real support	Pure twin	Fingerprint	+ physics
5	0.286	0.166	0.203	0.199
20	0.316	0.172	0.177	0.266
50	0.325	0.166	0.168	0.231

Three things are true in every row. Nothing beats simply training on the real labeled samples. The pure twin is the *worst* entry, below the chance level of 0.333, even though its physics matches the class definition exactly. And the real-sample baseline does not improve with k — 0.286 at five samples per class, 0.325 at fifty — so the ceiling here is set by the bearing-disjoint protocol, not by sample count. Within the augmentation columns the ordering is stable too: layering the impulse train onto the target envelope is the highest augmentation at $k = 20$ (0.266 against 0.177) and at $k = 50$ (0.231 against 0.168), and is level with the envelope alone at $k = 5$ (0.199 against 0.203). Both of those exceed the pure twin in all three rows.

We must be equally clear about what this table cannot do. With every entry at or below chance, it cannot rank methods, and the differences between the augmentation columns lie within one standard deviation. We report it for the one comparison that is stable across all three values of k, and that is externally corroborated.

A positive control for the mechanism. On CWRU, fingerprint calibration was the best augmentation and layering physics on it consistently hurt. On Paderborn the ranking inverts: fingerprint calibration alone is indistinguishable from the raw twin, and adding the physical impulse train is what recovers the ground it lost. This is exactly what the mechanism predicts. On CWRU the class information *is* the fingerprint, so estimating the fingerprint captures the label and physics is redundant noise. On Paderborn we measured that the class-mean fingerprints are identical across classes (margin +0.003, Table 4); calibrating to them therefore transfers no label information, and the only thing left carrying the label is the impulse repetition rate — the physics. The methodological failure we diagnose is thus not a property of

physics-based augmentation as such, but of applying it to a benchmark whose class labels are bound to individual recordings.

8. Method: Target-Domain Spectral Envelope Calibration

Let $\{(\mathbf{x}_i, y_i)\}_{i=1}^{kC}$ be the labeled target support set for C classes. For class c we estimate the magnitude envelope

$$A_c(f) \;=\; \frac{1}{|\mathcal{S}_c|} \sum_{i \in \mathcal{S}_c} \big|\mathcal{F}\{\mathbf{x}_i \odot \mathbf{w}\}(f)\big|, \tag{1}$$

smoothed with a moving average of width 5 bins and normalized to unit ℓ_2 norm. Every envelope number quoted in this paper (hereafter the *smoothed envelope*) uses this definition; the unsmoothed class-mean spectra of Table 4 are a different statistic and are not comparable to it. Synthetic samples are then obtained by coloring an excitation e in the frequency domain,

$$\mathbf{x}_{\text{syn}} = \mathcal{F}^{-1}\big\{\, \mathcal{F}\{e\} \cdot A_c \,\big\}, \tag{2}$$

followed by per-window z-scoring. We consider two excitations: white noise (the *fingerprint* variant) and an impulse train placed at the characteristic frequencies recomputed at the *target* shaft speed (the *fingerprint+physics* variant).

Both variants use only the k labeled support samples; no test window is involved. The envelopes are accurate at small k: estimated from $k = 5$ samples, the per-class envelope matches one estimated from the entire target load at cosine 0.995 on average (0.989 worst class). The corresponding *source-domain* envelopes match the target load only at 0.924, which is why the target-domain qualifier matters.

9. Results

9.1. Standard protocol

Table 8 reports the ten-class S0→S3 comparison, five seeds, all runs from a single uniform batch. Three observations.

The twin is worse than its own ablation. Adding the calibrated physics twin (DT_DANN) reaches macro-F1 0.914 ± 0.078, below plain adversarial adaptation (DANN, 0.986 ± 0.027) and below the physics-free augmentation ablation RAND_DANN (0.922 ± 0.098). It is also the least stable configuration.

Table 8: Ten-class CWRU S0→S3, $k = 5$ labeled target samples per class, five seeds, single uniform batch. SRC_PURE uses no target data at all; SOURCE_ONLY and all remaining rows additionally use the labeled support set.

Method	Accuracy	macro-F1
SRC_PURE (no target data)	0.8188 ± 0.0421	0.7307 ± 0.0225
SOURCE_ONLY	0.8842 ± 0.0377	0.8228 ± 0.0686
DANN (A2 ablation)	0.9904 ± 0.0179	0.9858 ± 0.0268
RAND_DANN (A3 ablation)	0.9512 ± 0.0553	0.9223 ± 0.0977
DT_DANN (physics twin)	0.9028 ± 0.1212	0.9141 ± 0.0780
FP_PHYS_DANN (fp. + imp.)	0.9917 ± 0.0091	0.9891 ± 0.0123
fp_dann (fp. + noise)	$\mathbf{0.9998 \pm 0.0003}$	$\mathbf{0.9998 \pm 0.0004}$

The protocol saturates. Every configuration that uses target labels reaches 0.91–1.00, leaving almost no headroom; the differences among DANN, FP_PHYS_DANN and FP_DANN are not a meaningful demonstration of method quality in this regime. The informative comparisons are in the constrained regimes of Section 9.

Performance is bounded by the labeled support set. The zero-label bound SRC_PURE reaches 0.731; merely adding the $k = 5$ labeled target samples (SOURCE_ONLY) raises this to 0.823.

9.2. Constrained regimes

The standard protocol saturates, so the meaningful comparisons are the constrained ones. In the synthesis-only regime (Table 1) the physics twin is at chance (0.123) while fingerprint-coloured white noise reaches 0.910; adding the physical impulse train drops this to 0.651, and removing the fault-harmonic fine structure from the envelope drops it further to 0.524.

In the no-source regime, where only the k labeled target samples are available (Table 9), the same ordering holds: the twin augmentation is harmful at every k, whereas spectral augmentation helps, and most at $k = 1$ ($0.890 \rightarrow 0.977$). Source-domain fingerprints, which need no target labels, remain clearly worse than target-domain ones at every k (0.861/0.905 against 0.977/0.993).

Table 9: No-source few-shot setting, target load 3, macro-F1. Each cell is 6 runs (three seeds, the whole configuration repeated twice). Augmentations add 400 samples per class generated from the k labeled support samples only.

Augmentation	$k = 1$	$k = 3$	$k = 5$
none (support only)	0.890	0.970	0.985
physics twin	0.558	0.777	0.885
fingerprint + white noise	0.977	0.995	0.993
fingerprint + impulse	0.947	0.983	0.658[†]
source fingerprint	0.861	0.945	0.905
phase surrogate	0.960	0.993	0.993
noise-only augmentation	0.952	0.991	0.994
true target samples (bound)	1.000	1.000	1.000

[†] very high variance (±0.46): two of six runs collapse to 0.014, well below the 0.10 chance level. The sample is bimodal rather than noisy.

10. Discussion and Limitations

Scope. The conclusion is emphatically *not* that simulation-based augmentation is useless in general. It is that the two benchmarks we examine cannot test it, for different reasons that share a root. On CWRU the class label is confounded with the recording, so the signal a classifier learns is not the fault. On Paderborn the label *is* physically meaningful — damage location, the very quantity whose characteristic frequencies a model can compute — yet the magnitude spectrum carries none of it, because a single rig tone and its spectral leakage hold over 99% of the energy in every bearing whether healthy or damaged. A benchmark suitable for testing physics-based augmentation must satisfy both conditions at once: the label must have a physical definition, and the recording must actually expose it. Neither of the two most-used public datasets does, and we suggest that establishing this be part of benchmark validation rather than something each paper discovers separately. The CWRU benchmark has already been criticised on related grounds [26]; we add a mechanism and a quantitative control. The situation is not unique to this field — a model can generalise across hospitals while relying on a site-specific confounder invisible in the acquisition protocol [27] — which is why we state the requirement in terms of benchmark construction rather than of any one paper's methodology.

This also reframes what the field should take from negative results. The failure we diagnose is a property of the *pairing* of a generator with a bench-

mark, not of the generator. The same twin that is worthless on CWRU is, on Paderborn, no longer the worst entry once its impulse train is layered onto the target spectral envelope (Table 7): that is the only setting in which we observe the physical component adding anything, and it is what the mechanism predicts, since there the fingerprint carries no label and the only remaining carrier of the label is the impulse repetition rate. The pure twin remains the worst augmentation on *both* benchmarks, and on neither does any augmentation beat simply training on the real labeled support samples.

Implications for reported numbers. Two papers reporting the same method on CWRU may differ by 0.1–0.3 macro-F1 purely because of how the few-shot split was drawn. Future work should state whether support and test come from the same recording, and ideally report both. The same control is cheap wherever a dataset contains several independent specimens per class: hold out specimens, not windows. Our own experience is a case in point. The secondary tables in this paper are reported over six, ten or twenty runs rather than the conventional three, because repeated execution of *the same command* with the same seeds moved a ten-seed mean by up to 0.033 and made a three-seed standard deviation wrong by more than an order of magnitude. This is the same problem that has been documented for reinforcement learning [28] and for evaluation budgets generally [29], and it argues for reporting score *distributions* and the compute spent on them.

A benchmark validity checklist. The failures above share a structure: a benchmark can make a positive or a negative result about simulation-derived samples uninterpretable before any method is run. Table 10 states the properties a benchmark must have, the check that establishes each, the signature of its failure, and where this paper observes that signature. The first two rows cost nothing beyond a second evaluation, the third is one FFT per class, and we suggest reporting all three in the paper rather than in a supplement. The fourth row is a different matter: Section 7.4 shows that filtering the dominant line out of the data does not remove the leakage, so observability and leakage have to be checked separately rather than one standing in for the other.

Limitations. (i) We study two datasets and one backbone. (ii) The cross-recording control on CWRU evaluates on a whole different load, which confounds the recording change with the operating-condition change; a same-load, different-session split is not available in CWRU, and we resort to Paderborn to separate the two terms of Section 7. There the separation relies on

Table 10: A benchmark validity checklist for simulation-based augmentation. Each row is a property a benchmark must have before a positive or a negative result about simulation-derived samples can be interpreted; the last column records where this paper observes the corresponding failure.

Dimension	Required check	Signature of failure	Observed here
Specimen independence	Each class contains independent bearings, and no specimen contributes to both support and test	One bearing reused across the split	CWRU: one bearing per class
Recording independence	Support and test from different recordings; report both numbers and the support-only baseline	A random window split of one recording, so that using any target label beats training on the support set	CWRU: +0.15–0.19
Spectral separability	Within- versus between-class mean spectra, in source and target	Between-class separation no larger than within-class	Paderborn: margin +0.003
Physical observability	Fault characteristic frequencies visible above the recording's own structure	A rig tone or recording fingerprint dominates the band	Paderborn: 100 Hz line, over 99% of the energy
Temporal scale	The window spans at least one characteristic period at the sampling rate	The window is shorter than one fault cycle	Paderborn: 1024 samples = 16 ms
Synthetic validity	Synthetic and real class-conditional spectra agree, and a discriminator cannot separate them	The generator reproduces no target class	CWRU: twin macro-F1 0.123

20 repeated measurements per bearing, so it measures the session term under one rig's repeat-measurement protocol and may not transfer to datasets whose repeated recordings differ by more than a re-mount. (iii) Our constructive method is a regularizer rather than a uniform improvement, and we report it as such. (iv) In the Paderborn subset we downloaded, every artificial-damage bearing is outer-ring and every run-to-failure bearing is inner-ring, so damage location and damage origin are confounded; our spectral claim is an *absence* of class structure and is unaffected, but it does not support a separate statement about outer versus inner race. (v) The Paderborn few-shot entries all lie at or below chance, so that table ranks nothing; we use it only for the two comparisons that are stable across all three sam-

ple budgets. (vi) The spectral comparison for Paderborn uses one operating condition and 21 bearings.

11. Conclusion

We showed that on CWRU cross-load bearing diagnosis the dominant class information is a per-bearing recording fingerprint that physics cannot predict, making simulation-based sample supplementation structurally ineffective on that benchmark; that the standard few-shot protocol leaks this fingerprint from support to test and inflates reported accuracy by 0.15–0.19, rising to 0.31 in the hardest direction; and that a compact spectral envelope calibration estimated from five labeled samples per class is the most robust augmentation available, while adding physical impulse structure on top of it consistently hurts.

We then removed the objection that CWRU is simply a badly chosen dataset. On Paderborn, where several independent bearings share each class, class-mean magnitude spectra of healthy, inner-ring and outer-ring bearings are indistinguishable (margin +0.003 after the rig tone is filtered out, against 0.626 on CWRU), so CWRU's class separation is a property of individual specimens rather than of the classes; same-recording evaluation remains optimistic, by +0.386, of which +0.161 is attributable to the recording session alone and +0.225 to the change of specimen — and removing the rig tone, which holds 93% of the spectral energy, leaves +0.321, still dominated by the specimen term, so the optimism is not an artefact of that line; and the physical component of the twin becomes useful for the first time once its impulse train is layered on the target envelope, in the one setting where the labels rather than the fingerprints carry the physics — though the pure twin is still the worst augmentation there, and no augmentation on either benchmark beats training on the real labeled support samples. The practical recommendation is therefore not to abandon physics-based generation, but to validate the benchmark before trusting either the positive or the negative result obtained on it.

CRediT authorship contribution statement

Fangyu Zhang: Conceptualization, Methodology, Software, Formal analysis, Investigation, Resources, Writing – original draft, Writing – review & editing, Project administration. **Julong He:** Supervision, Validation, Data curation, Visualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

The CWRU 12 kHz drive-end and Paderborn KAt datasets analysed here are public. The code and the per-table artifacts that reproduce every number in this paper are available at `https://github.com/qy321666/WSlB` (release `v1.8`) and are also provided as supplementary material to this submission.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the authors used ClawsGO Science Agent in order to improve the language of the manuscript. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

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